PHYS4055
Atomic Physics & Quantum Optics

Week 1, Lecture 1

Acknowledgement of Country

I would like to begin by acknowledging the Traditional Owners and their custodianship of the lands on which we meet.

We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

We recognise their valuable contributions to Australian and global society.



PHYS4055
Course overview

Who am I?



Dr Josephine Dias
Research Fellow

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Office: Parnell Room 320
Office hours: Thursdays 3-4 pm

Research interests: quantum information, quantum communication (theory)



Fun fact about me:
I used to live and work in Japan



Tokyo



Okinawa

This is a space for *everyone*

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The School of Mathematics and Physics values inclusivity

It is the responsibility of everyone in SMP to ensure that classrooms and workspaces are caring, inclusive, and empowering environments for all students and staff.

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- **Be an ethical bystander.**
- **Act if you experience or witness bullying, harassment or discrimination.**
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SMP Inclusivity Champions (updated 22/07/2026)

To email: Click on the hyperlink, or add uq.edu.au to the address

UQ ALLY

Every cohort has a nominated contact – however, you are welcome to contact whomever you choose.

Our champions are committed to listening to your inclusivity concerns and ensuring they are suitably addressed.



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MATH1061/81
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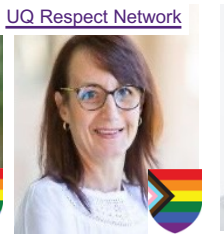


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Head of School
Joseph Grotowski
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Structure of this course

Four modules, roughly 3 weeks each:

1. Quantum optics theory
2. Quantum optics applications
3. Atomic physics theory
4. Atomic physics applications

Optics lecturers:



Dr Josephine Dias



Dr Laura Serino

Atomic physics lecturer:



Associate Professor Jacinda Ginges

Assessment

Tutorial/ Problem Set	Problem set	25%	14/08/2026 4:00 pm 4/09/2026 4:00 pm 25/09/2026 4:00 pm 23/10/2026 4:00 pm
Presentation	Course Seminar Identity Verified In-person	15%	Seminars will be scheduled during regular lecture times between Week 10 and Week 11 inclusive. Dates will be given in the first week of the course.
Examination	End of Semester Exam Hurdle Identity Verified In-person	60%	End of Semester Exam Period 7/11/2026 - 21/11/2026

Problem Set

Each problem set covers one module of the course

Tutorial/ Problem Set	Problem set	25%	14/08/2026 4:00 pm	1. Quantum optics theory
			4/09/2026 4:00 pm	2. Quantum optics applications
			25/09/2026 4:00 pm	3. Atomic physics theory
			23/10/2026 4:00 pm	4. Atomic physics applications

First problem set on quantum optics theory will be on blackboard **31/07**

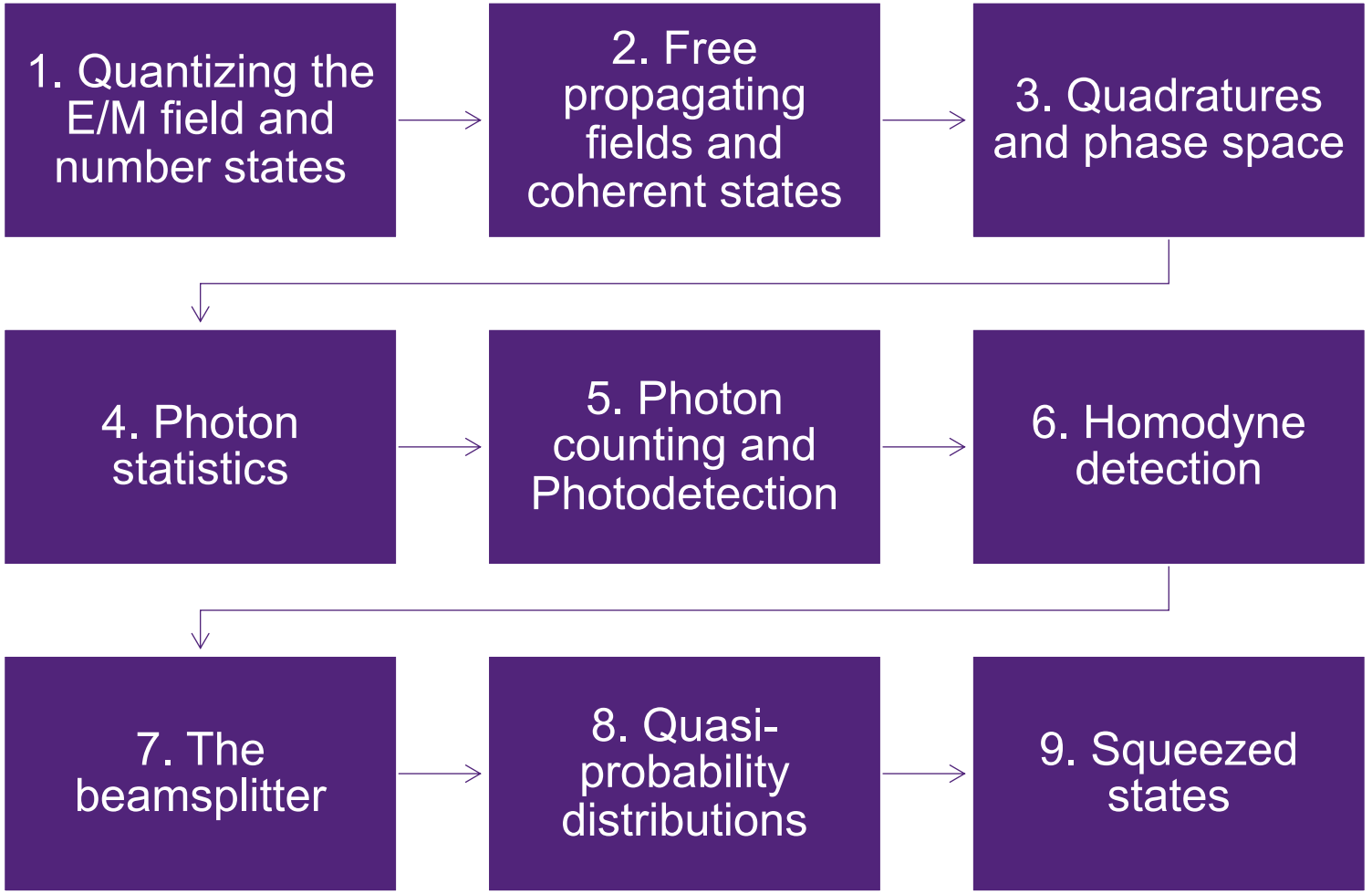
Seminar

- Each student will deliver a 10-minute seminar on a topic related to the course: an 8-minute presentation followed by 2 minutes of questions.
- Seminars will follow a conference-style format, with presentations delivered back-to-back.
- A list of suggested topics will be provided, but students may also propose their own topic.
- All presentations will be delivered in person.

Dates: **Week 10 Thursday 8/10**

Week 11 Tuesday 13/10 and Thursday 15/10

What we will cover – Quantum Optics Theory



Quantum optics applications

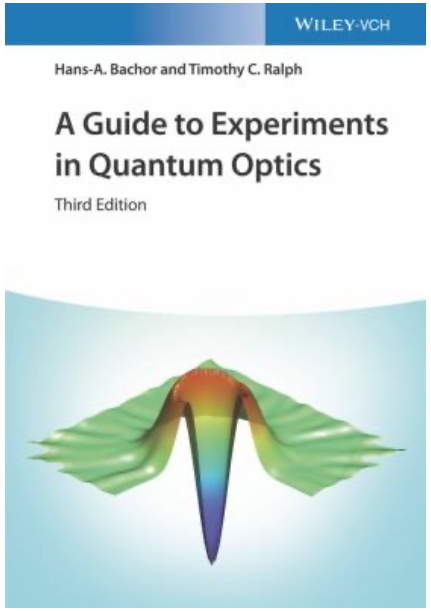
Topics we will cover later in the semester

- Quantum metrology
- Optical quantum information
- Quantum communication
- Entanglement

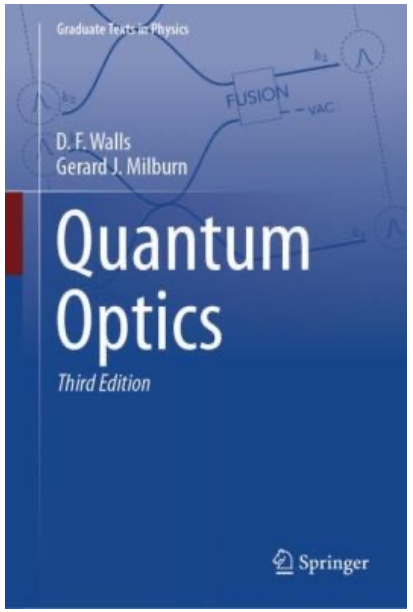
And more…

Quantum optics resources

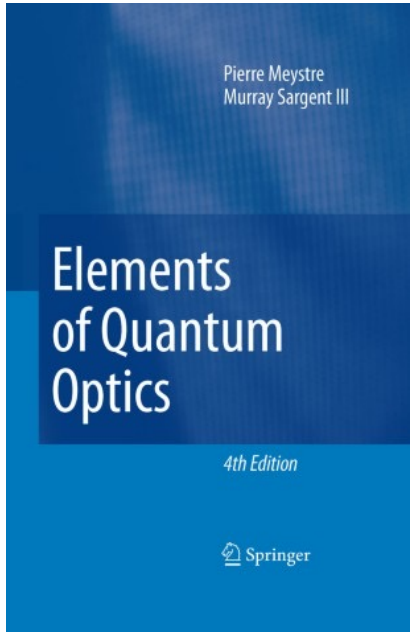
Textbooks



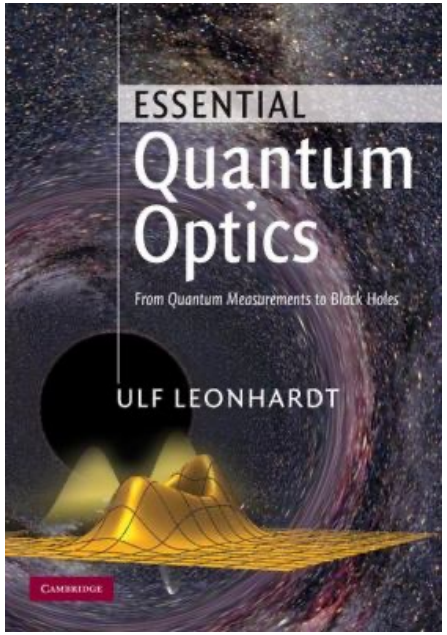
[A guide to experiments in quantum optics / Hans-A. Bachor and Timothy C. Ralph]
Bachor, H.-A. (Hans-Albert), author
Third edition 2019



[Quantum Optics / by D. F. Walls, Gerard J. Milburn]
Walls, D. F
3rd ed. 2025



[Elements of Quantum Optics]
Meystre, Pierre
4th ed. 2007. 2007



[Essential Quantum Optics: From Quantum Measurements to Black Holes]
Leonhardt, Ulf
1 2010

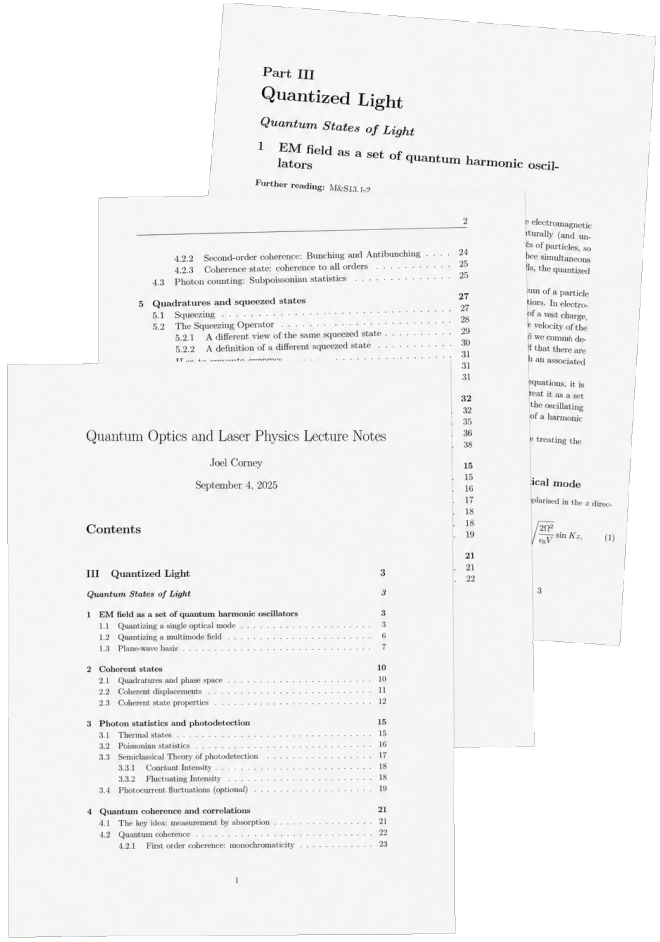
All available online

Quantum optics resources

Lecture notes

These lecture notes cover the introductory quantum optics material taught in Weeks 1–3.

Available via Blackboard



Staying on top of content

Readings:

specified in lecture slides/quantum optics notes

Consultation hour:

Thursday 3-4pm in my office 07-320,

available during Weeks 1-6, otherwise by appointment.

Ed discussion board:

a good place for students to discuss and clarify

Academic integrity:

acknowledge all sources; some discussion among peers is fine, though no collusion. Your work needs to be your own



Week 1, Lecture 1
Quantizing the electromagnetic field

Lecturer: Josephine
Email: j.dias@uq.edu.au
Office: Parnell Room 320
Office hours: Thursdays 3-4 pm

Notes 1.1, 1.2; Bachor & Ralph Ch 4.1, 4.2; Walls and Milburn Ch 2.1, 2.2

Where are we going?



Classical light



Experimental problems



Electromagnetic modes

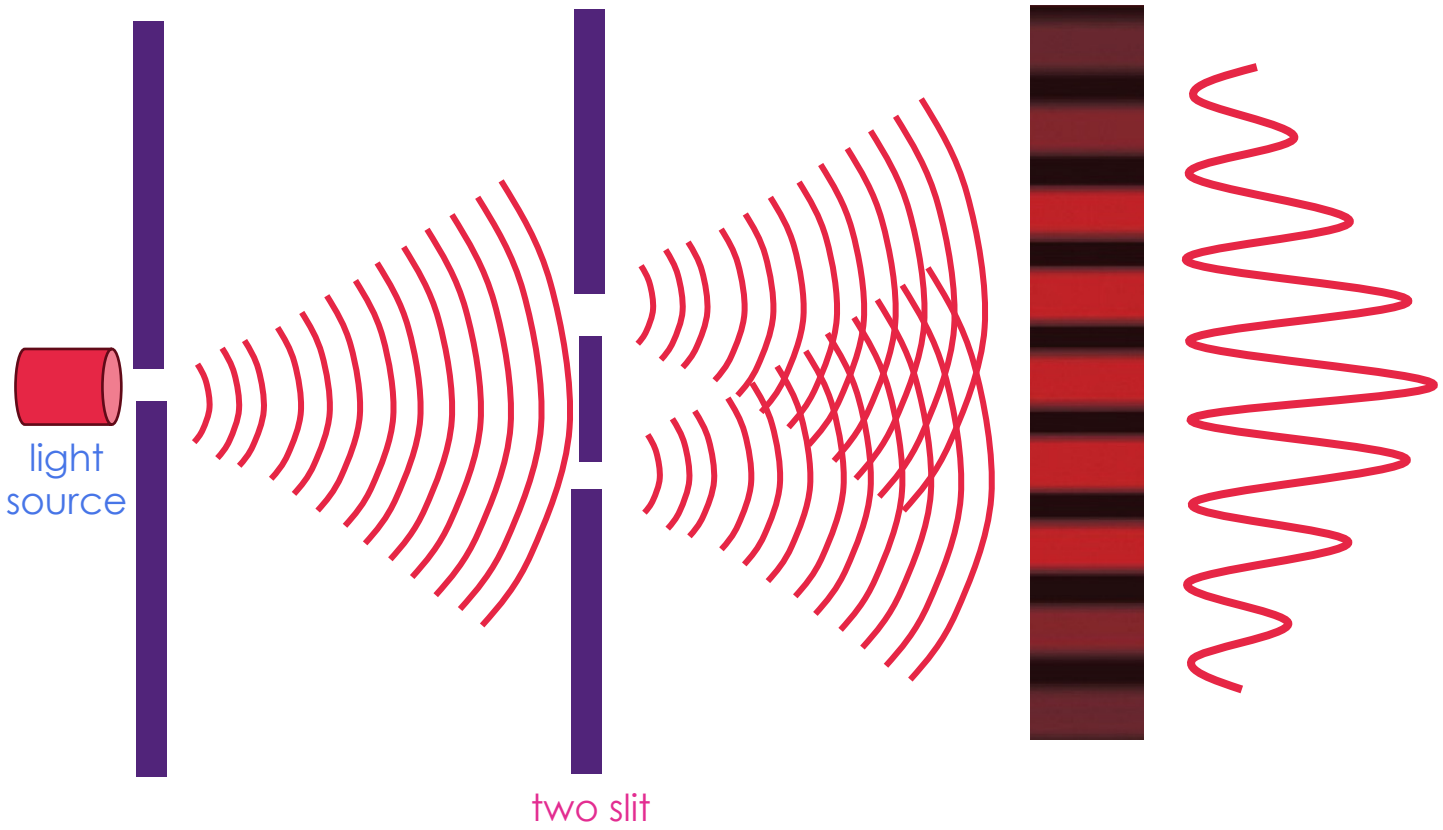


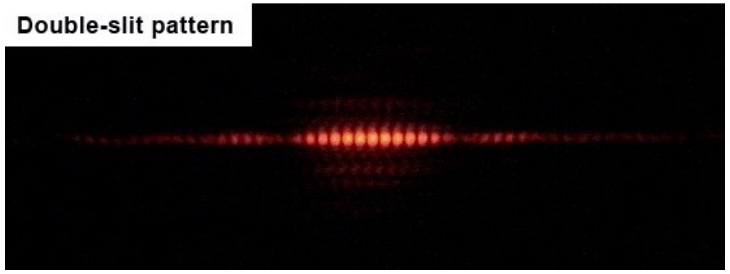
Harmonic oscillator



Photons

Classical Light





Classical Light

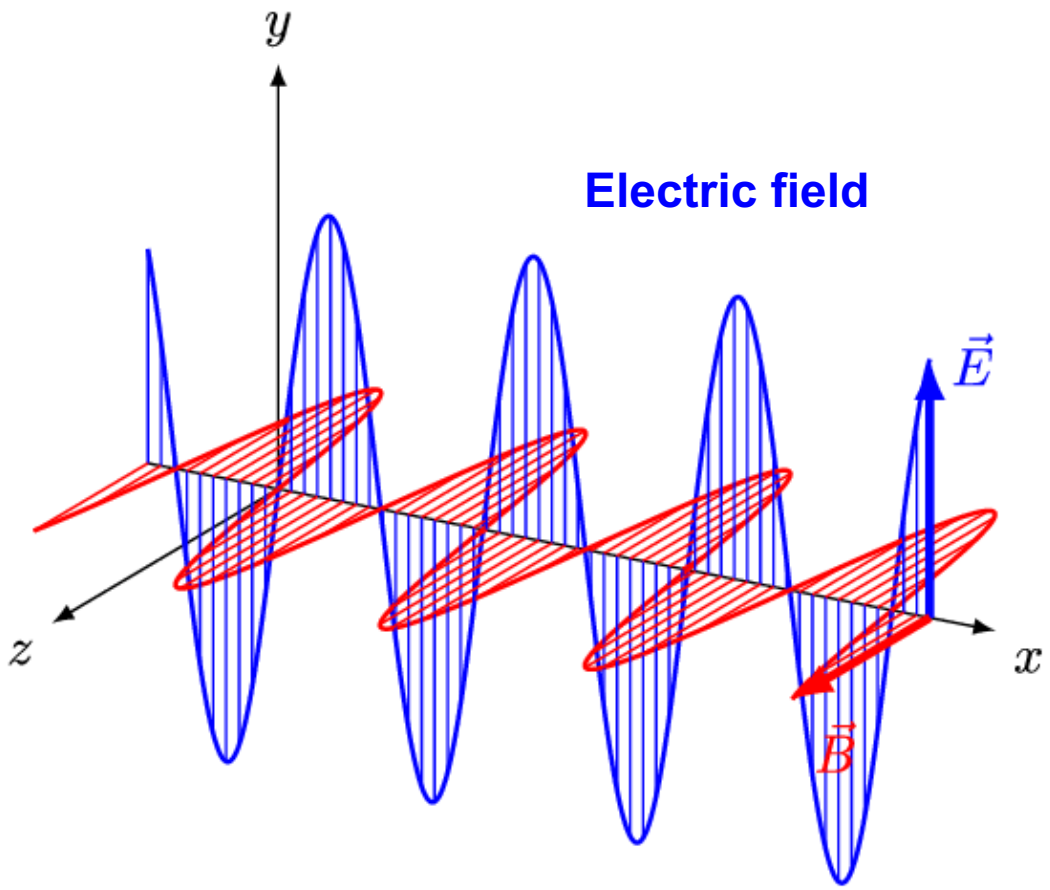
Maxwell's equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0},$$

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$



Why do we need to quantize light?

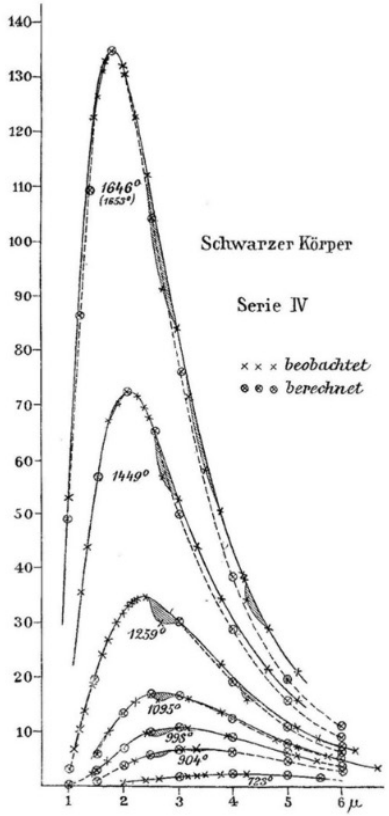
Black-body radiation: the first step towards quantized light

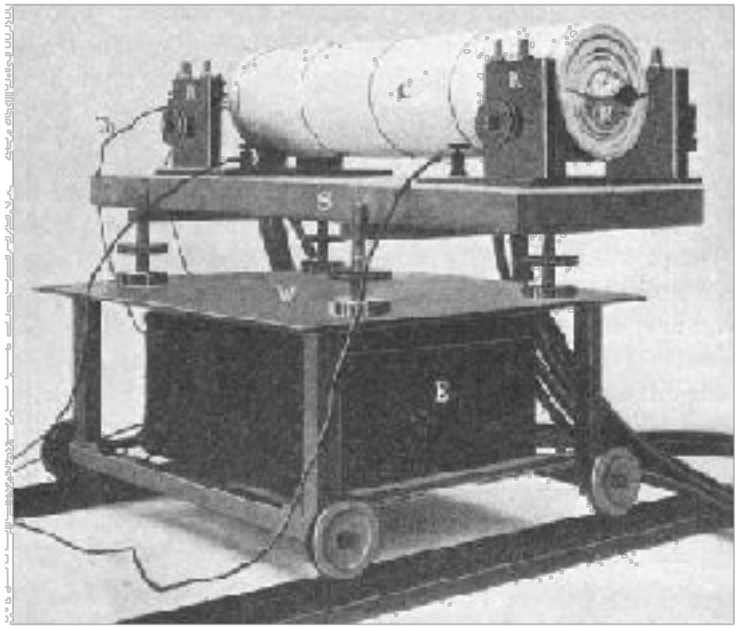
In 1900 Planck (motivated by this experiment by Lummer and Pringsheim) found that black-body radiation could only be correctly modelled by assuming that atomic light emission was quantized

Planck reproduced the observations by assuming that electromagnetic oscillators exchange energy in discrete amounts:

$$E_n = n\hbar\omega.$$

This assumption marked the beginning of quantum theory.



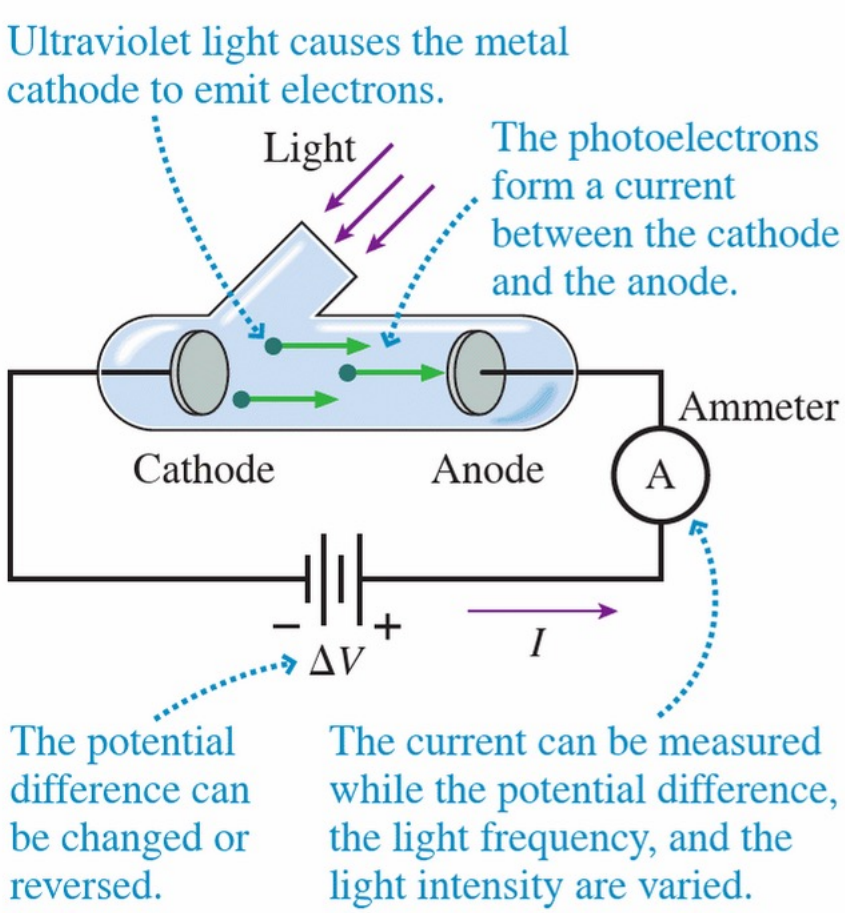


Why do we need to quantize light?

The photoelectric effect: light exchanges energy in packets

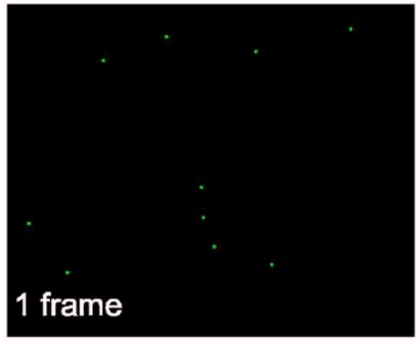
- Electrons are emitted only above a threshold frequency.
- Electron energy increases with the frequency of the light.
- Increasing intensity increases the number of emitted electrons, not their maximum energy.

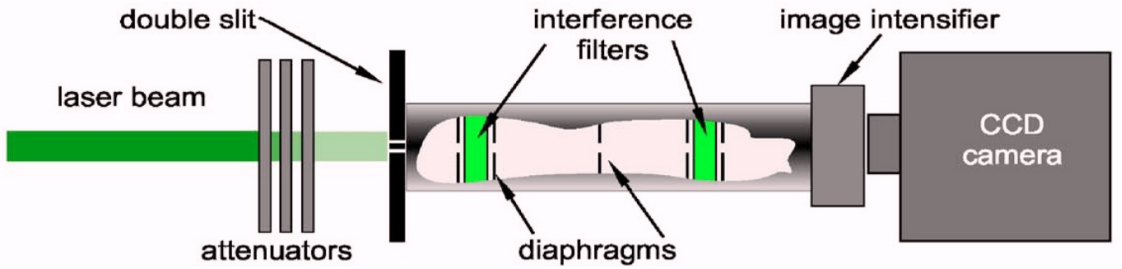
Phet Simulation



Double slit experiment with photons

Each dot represents one photon





T. L. Dimitrova, A. Weis; The wave-particle duality of light: A demonstration experiment. *Am. J. Phys.* 1 February 2008; 76 (2): 137–142. https://doi.org/10.1119/1.2815364

Development of the laser



What does it mean to quantize something?

Canonical quantization

Classical description

$x(t), p(t)$ are numbers

$$H = \frac{p^2}{2m} + V(x)$$

Quantum mechanics

$$x, p \longrightarrow \hat{x}, \hat{p}$$

Classical Poisson bracket becomes quantum commutator

$$[\hat{x}, \hat{p}] = i\hbar$$

Classical Hamiltonian becomes a quantum operator

$$H(x, p) \longrightarrow \hat{H}(\hat{x}, \hat{p})$$

Time indep. Schrodinger equation

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

Quantizing light means quantizing its modes

An electromagnetic field has an electric field $\mathbf{E}(\mathbf{r}, t)$and a magnetic field $\mathbf{B}(\mathbf{r}, t)$ at every position $\mathbf{r}$.

$$\mathbf{E}(\mathbf{r}, t) = \sum_j q_j(t)\mathbf{u}_j(\mathbf{r})$$

$u_j(\mathbf{r}) = \text{spatial profile of mode } j$

$q_j(t) = \text{time-dependent amplitude of mode } j$

$\omega_j = \text{mode frequency}$

One electromagnetic mode ⟷ One harmonic oscillator

Maxwell's equations in free space

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0},$$ Gauss's law

$$\nabla \cdot \mathbf{B} = 0,$$ Gauss's law for magnetism

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$ Faraday's law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$ Ampère-Maxwell law

In free space (containing no charges or currents),

$$\rho = 0, \qquad \mathbf{J} = 0$$

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = 0$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t}$$

$$\nabla \times \mathbf{B}(\mathbf{r}, t) = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t}$$

Introducing electromagnetic potentials

$$\nabla \cdot \mathbf{E}(\mathbf{r},t) = 0$$
$$\nabla \cdot \mathbf{B}(\mathbf{r},t) = 0$$
$$\nabla \times \mathbf{E}(\mathbf{r},t) = -\frac{\partial \mathbf{B}(\mathbf{r},t)}{\partial t}$$
$$\nabla \times \mathbf{B}(\mathbf{r},t) = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}(\mathbf{r},t)}{\partial t}$$

Introduce the scalar and vector potentials:

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}} \qquad \boxed{\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}}$$

Vector identities

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$
$$\nabla \times (\nabla \phi) = 0$$

Coulomb gauge

The electromagnetic potentials are not unique. We choose the Coulomb gauge:

$$\boxed{\nabla \cdot \mathbf{A} = 0}$$

Using $\nabla \cdot \mathbf{E} = 0$ and $\mathbf{E} = -\nabla\phi - \dfrac{\partial \mathbf{A}}{\partial t}$

We obtain $-\nabla^2\phi - \dfrac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = 0$

In the Coulomb gauge, $\nabla^2\phi = 0$

For a source-free radiation field, we may choose $\boxed{\phi = 0}$

and therefore

$$\boxed{\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}}$$

Wave equation for the vector potential

Start from the Ampère–Maxwell law: $\nabla \times \mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$

Substitute $\mathbf{B} = \nabla \times \mathbf{A}, \qquad \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$

This gives $\nabla \times (\nabla \times \mathbf{A}) = -\frac{1}{c^2}\frac{\partial^2 \mathbf{A}}{\partial t^2}$

Use the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$

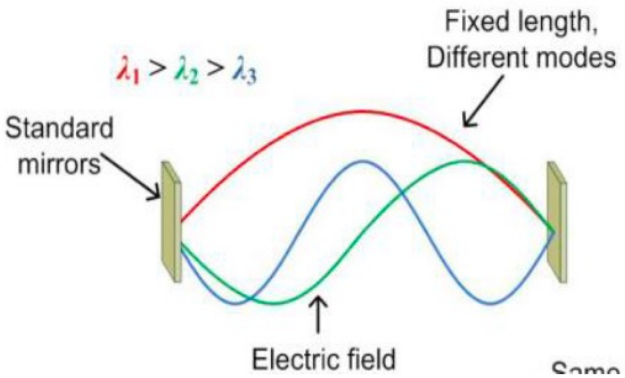
In the Coulomb gauge, $\nabla \times (\nabla \times \mathbf{A}) = -\nabla^2 \mathbf{A}$

Therefore, the vector potential obeys the wave equation $\boxed{\nabla^2 \mathbf{A}(\mathbf{r},t) - \frac{1}{c^2}\frac{\partial^2 \mathbf{A}(\mathbf{r},t)}{\partial t^2} = 0}$

And because $\boxed{\mathbf{B} = \nabla \times \mathbf{A}}$ and $\boxed{\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}}$

The E and B fields also obey the wave equation

Electromagnetic field in a cavity



Consider a cavity with;
- Mode volume V.
- Single mode.
- Monochromatic.
- Single polarization.

Seek separable solutions of the form

$$\mathbf{A}(\mathbf{r},t) = q_j(t)\mathbf{u}_j(\mathbf{r})$$

For a field polarised along the x-direction and propagating along z,

$$\mathbf{u}_j(\mathbf{r}) = \hat{\mathbf{x}}\, C_j \sin(k_j z)$$

The boundary condition at $z = L$ requires

$$\sin(k_j L) = 0$$

Therefore,

$$\boxed{k_j = \frac{j\pi}{L}, \qquad j = 1, 2, 3, \ldots}$$

The corresponding angular frequencies are

$$\boxed{\omega_j = ck_j = \frac{j\pi c}{L}}$$

Select a single cavity mode

Choose one mode of angular frequency $\omega$and omit the mode index:

$$\boxed{\mathbf{A}(\mathbf{r},t) = q(t)\mathbf{u}(\mathbf{r})}$$

For a cavity of volume V, choose

$$\mathbf{u}(\mathbf{r}) = \hat{\mathbf{x}}\sqrt{\frac{2}{\varepsilon_0 V}}\sin(kz)$$

This mode is normalized according to

$$\boxed{\int_V \varepsilon_0 \left|\mathbf{u}(\mathbf{r})\right|^2 \, \mathrm{d}^3\mathbf{r} = 1}$$

Substitute the single-mode form into the wave equation:

$$q(t)\nabla^2\mathbf{u}(\mathbf{r}) - \frac{1}{c^2}\ddot{q}(t)\mathbf{u}(\mathbf{r}) = 0$$

Using $\nabla^2\mathbf{u}(\mathbf{r}) = -k^2\mathbf{u}(\mathbf{r})$

we obtain

$$\boxed{\ddot{q}(t) + \omega^2 q(t) = 0, \qquad \omega = ck}$$

Energy of the single mode

The fields are $\mathbf{E}(\mathbf{r},t) = -\dot{q}(t)\mathbf{u}(\mathbf{r})$ $\mathbf{B}(\mathbf{r},t) = q(t)\nabla \times \mathbf{u}(\mathbf{r})$

The classical electromagnetic-field energy is

$$H = \frac{1}{2}\int_V \left[\varepsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0}|\mathbf{B}|^2\right] \mathrm{d}^3\mathbf{r}$$

Using the mode normalization gives

$$\boxed{H = \frac{1}{2}\dot{q}^2 + \frac{1}{2}\omega^2 q^2}$$

Define the conjugate momentum $p = \dot{q}$

The Hamiltonian becomes

$$\boxed{H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2}$$

Canonical quantization of the mode

Promote the classical variables to operators:

$$q \longrightarrow \hat{q}, \qquad p \longrightarrow \hat{p}$$

Impose the canonical commutation relation:

$$\boxed{[\hat{q}, \hat{p}] = i\hbar}$$

The Hamiltonian becomes

$$\boxed{\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2}$$

Creation and annihilation operators

Define the annihilation operator:

$$\hat{a} = \sqrt{\frac{\omega}{2\hbar}}\,\hat{q} + \frac{i}{\sqrt{2\hbar\omega}}\,\hat{p}$$

Define the creation operator:

$$\hat{a}^\dagger = \sqrt{\frac{\omega}{2\hbar}}\,\hat{q} - \frac{i}{\sqrt{2\hbar\omega}}\,\hat{p}$$

They satisfy

$$[\hat{a}, \hat{a}^\dagger] = 1$$

The inverse relations are:

$$\hat{q} = \sqrt{\frac{\hbar}{2\omega}}\,(\hat{a} + \hat{a}^\dagger) \qquad \hat{p} = -i\sqrt{\frac{\hbar\omega}{2}}\,(\hat{a} - \hat{a}^\dagger)$$

The electric-field operator

For a single standing-wave cavity mode polarised in the x-direction,

$$\hat{\mathbf{E}}(z,t) = \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}}\left[\hat{a}(t) + \hat{a}^\dagger(t)\right]\sin(Kz)\,\hat{\mathbf{x}}$$

The operators evolve according to

$$\hat{a}(t) = \hat{a}e^{-i\omega t}, \qquad \hat{a}^\dagger(t) = \hat{a}^\dagger e^{i\omega t}$$

Therefore,

$$\hat{\mathbf{E}}(z,t) = \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}}\left(\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}\right)\sin(Kz)\,\hat{\mathbf{x}}$$

Classically,

$$\mathbf{E}(\mathbf{r},t) = i\omega\left(Qe^{-i\omega t} - Q^* e^{i\omega t}\right)\mathbf{u}(\mathbf{r})$$

Separate the field into positive- and negative-frequency components:

$$\hat{\mathbf{E}}(z,t) = \hat{\mathbf{E}}^{(+)}(z,t) + \hat{\mathbf{E}}^{(-)}(z,t)$$

The components satisfy

$$\hat{\mathbf{E}}^{(-)}(z,t) = \left[\hat{\mathbf{E}}^{(+)}(z,t)\right]^\dagger$$

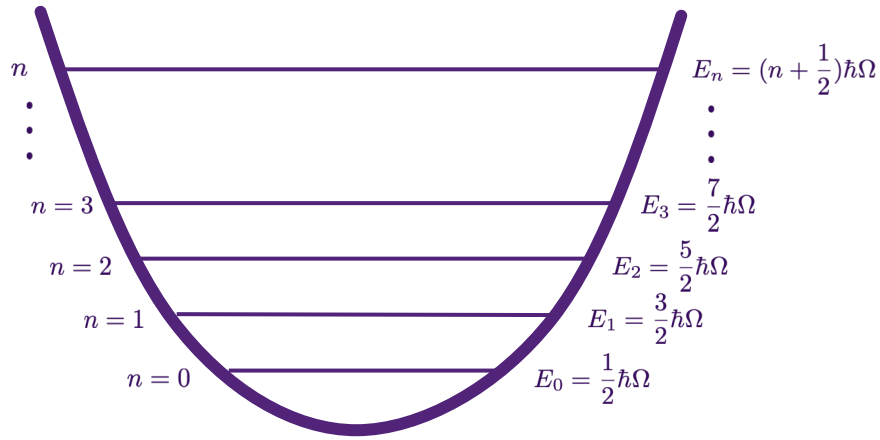
Photons as cavity-mode excitations

Define the number operator:

$$\hat{n} = \hat{a}^\dagger \hat{a}$$

The Hamiltonian is

$$\boxed{\hat{H} = \hbar\omega\left(\hat{n} + \frac{1}{2}\right)}$$



The number states satisfy

$$\hat{n}|n\rangle = n|n\rangle, \qquad n = 0, 1, 2, \ldots$$

The allowed energies are

$$\boxed{E_n = \hbar\omega\left(n + \frac{1}{2}\right)}$$

Field moments in a photon-number state

Expectation value

$$\boxed{\langle n|\hat{E}(z,t)|n\rangle = 0}$$

$$\begin{aligned}\langle \hat{E}^2(z,t)\rangle &= \langle n|\hat{E}^2(z,t)|n\rangle \\ &= \frac{\hbar\omega}{\varepsilon_0 V}\langle n|\left[\hat{a}(t)+\hat{a}^\dagger(t)\right]^2|n\rangle \sin^2(Kz)\end{aligned}$$

$$\boxed{\langle \hat{E}^2(z,t)\rangle = \frac{\hbar\omega}{\varepsilon_0 V}(2n+1)\sin^2(Kz)}$$

Electric-field variance

$$(\Delta E)^2 = \langle \hat{E}^2\rangle - \langle \hat{E}\rangle^2$$

$$\boxed{(\Delta E)^2 = \frac{\hbar\omega}{\varepsilon_0 V}(2n+1)\sin^2(Kz)}$$

Averaged over volume

$$\boxed{\overline{(\Delta E)^2} = \frac{\hbar\omega}{\varepsilon_0 V}\left(n+\frac{1}{2}\right)}$$

Photon-number states and the number operator

Number operator is Hermitian ⇒ Eigenvalues are real $\hat{n} = \hat{a}^\dagger \hat{a}$

Photon-number states are eigenstates of $\hat{n}$:

$$\boxed{\hat{n}|n\rangle = n|n\rangle, \qquad n = 0, 1, 2, \ldots}$$

The ladder operators act according to

$$\boxed{\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle}$$

$$\boxed{\hat{a}^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle}$$

The state $|0\rangle$ is the vacuum state and contains no photons in the selected mode. $\hat{a}|0\rangle = 0$

The state $|1\rangle$ contains exactly one photon, while $|n\rangle$ contains exactly n photons.

$$\boxed{|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle}$$

Constructing the first photon-number states

The vacuum state is $|0\rangle$ It satisfies $\hat{a}|0\rangle = 0$

One-photon state $\hat{a}^\dagger|0\rangle = \sqrt{1}\,|1\rangle = |1\rangle$

$$\boxed{|1\rangle = \hat{a}^\dagger|0\rangle}$$

Two-photon state

Apply the creation operator again:

$$\hat{a}^\dagger|1\rangle = \sqrt{2}\,|2\rangle$$

$$(\hat{a}^\dagger)^2|0\rangle = \sqrt{2}\,|2\rangle$$

$$\boxed{|2\rangle = \frac{1}{\sqrt{2}}(\hat{a}^\dagger)^2|0\rangle}$$

Three-photon state

Apply the creation operator once more:

$$\hat{a}^\dagger|2\rangle = \sqrt{3}\,|3\rangle$$

Starting from the vacuum,

$$(\hat{a}^\dagger)^3|0\rangle = \sqrt{1}\sqrt{2}\sqrt{3}\,|3\rangle = \sqrt{6}\,|3\rangle$$

$$\boxed{|3\rangle = \frac{1}{\sqrt{6}}(\hat{a}^\dagger)^3|0\rangle}$$

General result

$$\boxed{|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle}$$

Photon-number basis

The number states are orthonormal:

$$\langle m|n\rangle = \delta_{mn}$$

They form a complete basis:

$$\sum_{n=0}^{\infty} |n\rangle\langle n| = \hat{I} \qquad \{|n\rangle\} = \{|0\rangle, |1\rangle, |2\rangle, \cdots\}$$

Any state of a single optical mode can be expanded as

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle \qquad \sum_{n=0}^{\infty} |c_n|^2 = 1$$

A number state has no photon-number uncertainty:

$$\boxed{(\Delta n)^2 = \langle n|\hat{n}^2|n\rangle - \langle n|\hat{n}|n\rangle^2 = 0}$$

Summary

Experiments showed that classical physics cannot fully explain light.

Classically, light is described using electric and magnetic fields that obey Maxwell's equations.

Boundary conditions in an optical cavity select a discrete set of electromagnetic modes.

Each mode behaves like a harmonic oscillator.

The excitations of a quantized electromagnetic mode are photons:

$$\hat{H} = \hbar\omega\left(\hat{n} + \frac{1}{2}\right), \qquad \hat{n} = \hat{a}^\dagger\hat{a}$$